

BALLYPATRICK FOREST, United KingdomWMO: **039160**Lat: **55.1806N**Lon: **6.1547W**Elev: **156**StdP: **99.46**Time Zone: **0.00 (EUW)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	-0.3	-4.5	2.6	1.1	-3.3	2.9	1.5	14.8	5.5	13.5	5.2	2.6	210	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.7	20.6	16.6	19.1	15.8	17.8	15.0	17.3	19.6	16.4	18.4	15.5	17.3	4.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.8	18.5	15.4	11.1	17.4	14.6	10.6	16.5	49.3	19.6	46.4	18.5	43.9	17.4	20.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	10.5	9.3	-3.4	23.5	1.4	1.4	-4.4	24.5	-5.2	25.3	-5.9	26.0	-6.9	27.0		
(5)			-3.7	19.0	1.4	0.9	-4.7	19.7	-5.5	20.2	-6.3	20.7	-7.3	21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	4.7	4.8	5.7	7.3	9.6	12.0	13.6	13.6	12.3	9.8	7.0	5.2
	DBStd	4.09	2.45	2.55	2.54	2.52	2.61	2.36	2.18	1.88	2.07	2.33	2.48	2.74
	HDD10.0	855	163	146	136	89	38	6	0	0	3	32	94	148
	HDD18.3	3469	421	378	393	332	269	189	147	147	182	264	341	406
	CDD10.0	430	0	1	2	7	27	66	112	112	72	26	4	1
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	4	0	0	0	0	0	2	2	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.3	5.1	5.0	4.9	4.4	4.1	3.8	3.5	3.6	3.9	4.5	4.6	5.0
Precipitation	PrecAvg	1270	129	100	92	83	80	87	91	99	99	136	140	134
	PrecMax	1525	231	211	214	143	209	218	161	271	203	265	244	284
	PrecMin	959	25	30	28	22	14	27	34	21	8	52	73	69
	PrecStd	151	45	48	42	33	42	45	33	51	46	57	46	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	12.6	13.8	16.6	20.5	22.3	23.1	21.9	19.9	16.5	14.1	12.7
		MCWB	10.7	10.5	10.1	12.1	14.8	16.4	17.9	18.1	16.7	14.4	12.9	11.5
	2%	DB	10.6	11.2	11.8	14.4	18.0	19.4	20.7	19.8	17.9	15.2	12.7	11.5
		MCWB	9.7	9.7	9.6	11.1	13.7	15.3	16.9	16.8	15.5	13.6	11.7	10.5
	5%	DB	9.6	10.0	10.6	12.6	15.9	17.6	19.0	18.2	16.6	14.2	11.6	10.5
		MCWB	8.8	8.8	8.8	10.2	12.5	14.3	16.0	15.7	14.6	12.9	10.7	9.5
	10%	DB	8.7	8.9	9.5	11.2	14.1	16.0	17.5	17.0	15.5	13.3	10.7	9.4
		MCWB	7.8	7.7	8.0	9.1	11.4	13.5	15.0	14.9	13.7	12.1	9.8	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.0	11.4	13.1	15.7	17.4	18.7	18.6	17.5	15.0	13.2	12.0
		MCDB	11.5	12.0	12.9	15.4	19.2	20.6	21.8	21.0	19.1	16.0	13.9	12.7
	2%	WB	9.9	9.9	10.0	11.6	14.1	15.8	17.4	17.2	16.0	14.0	11.8	10.7
		MCDB	10.6	10.9	11.1	13.8	17.1	18.7	20.0	19.0	17.6	14.8	12.5	11.4
	5%	WB	8.9	8.9	9.1	10.4	12.9	14.7	16.3	16.2	14.9	13.1	10.9	9.7
		MCDB	9.5	9.9	10.3	12.3	15.5	17.0	18.5	17.8	16.2	14.0	11.5	10.4
	10%	WB	7.8	7.8	8.2	9.3	11.7	13.7	15.3	15.2	14.0	12.3	9.9	8.5
		MCDB	8.6	8.8	9.4	11.0	13.7	15.6	17.1	16.6	15.2	13.1	10.6	9.3
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	5.1	5.6	6.4	6.0	5.7	5.5	5.2	4.6	4.1	4.0
		MCDBR	5.1	5.3	6.1	7.4	8.9	8.4	8.0	7.4	6.4	5.3	4.7	5.0
		MCWBR	5.1	4.9	4.6	4.9	5.6	5.4	5.2	5.1	4.7	4.4	4.6	5.0
	5% WB	MCDBR	5.1	5.3	5.6	6.8	8.3	7.9	7.5	6.7	5.8	5.0	4.7	5.0
		MCWBR	5.1	5.1	4.7	4.9	5.7	5.6	5.2	5.1	4.8	4.6	4.8	5.1
Clear-Sky Solar Irradiance	taub	0.328	0.340	0.360	0.373	0.376	0.370	0.372	0.358	0.353	0.339	0.322	0.316	
	taud	2.243	2.336	2.335	2.337	2.373	2.421	2.417	2.472	2.480	2.449	2.309	2.203	
	Ebn at Noon	578	731	806	849	867	876	866	859	812	730	588	494	
	Edh at Noon	59	77	96	109	112	108	106	95	83	68	55	49	
All-Sky Solar Radiation	RadAvg	0.52	1.20	2.31	3.79	4.86	4.90	4.59	3.73	2.58	1.43	0.66	0.38	
	RadStd	0.06	0.11	0.19	0.29	0.43	0.49	0.37	0.29	0.16	0.14	0.07	0.04	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page